



It was in Coalbrookdale in Shropshire, England, that engineer Abraham Darby built the first coke-fired blast furnace to cast iron.

1.707 BILLION TONS THE AMOUNT OF IRON PRODUCED EACH YEAR AROUND THE WORLD

THE HUMAN PULSE WAS KNOWN AS AN INDICATOR OF HEALTH

more than 2,500 years ago. But it was not until English physician **John Floyer** invented his **pulse watch** in 1707 that western physicians began to measure the pulse in terms of heartbeats per minute. Floyer's timepiece was a watch that ran for exactly a minute while the physician counted pulses.

The following year, Dutch botanist and physician

Herman Boerhaave

developed a systematic approach to **diagnosis** that involved considering the patient's history, conducting a physical examination at the bedside, taking the pulse, and studying excretions.

Also in 1708, German physician, mathematician, and experimenter **Ehrenfried Walther von Tschirnhaus** (1651–1708)

discovered that he could make **porcelain** with a paste mixed from clay, alabaster, and calcium sulfate. Although the



Chinese had been making fine porcelain for centuries, the technology had eluded the west until this time.

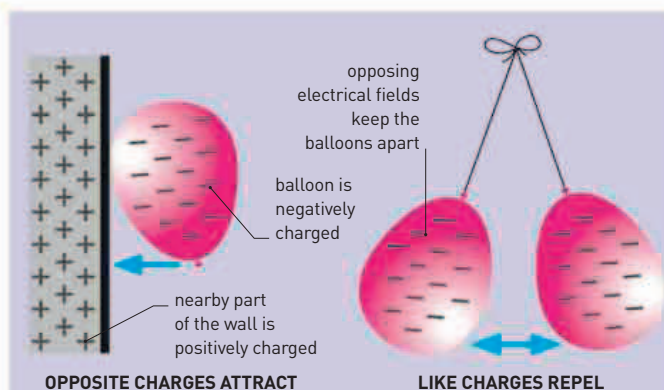
In 1709, English experimenter **Francis Hauksbee** published *Physico-Mechanical Experiments on Various Subjects*, in which he described his celebrated **experiments with static electricity**. Hauksbee discovered that by rubbing glass, he could

create static electricity and produce astounding

electrical effects, such as "electric light" (the glow inside a rotating evacuated glass sphere when rubbed), electric wind (the prickling sensation when rubbed glass is brought near the face), and electric repulsion and attraction.

English engineer **Abraham Darby**

Alcohol thermometer
Gabriel Fahrenheit's 1709 thermometer was the first compact device of its kind. It showed temperature by the expansion of colored alcohol. Later, versions using mercury were popular.



OPPOSITE CHARGES ATTRACT

LIKE CHARGES REPEL

STATIC ELECTRICITY

Static electricity is the build-up or deficiency of electrons (particles contained in atoms). Surfaces charged with excess electrons are attracted to surfaces that have lost electrons. Experiments to create static electricity were widely practiced in the 18th century, often with striking results. Some of the most important investigations were carried out by English experimenter Francis Hauksbee.

(1678–1717) revolutionized ironmaking in 1709 by **producing cast iron** in a coke-fueled blast furnace at Coalbrookdale in England. For the first time, iron could be cast in very large shapes, paving the way for the machines and engineering feats of the **Industrial Revolution**.

In Amsterdam, also in 1709, Polish-Dutch physicist **Gabriel Daniel Fahrenheit** (1686–1736) constructed an **alcohol-filled thermometer**. It was the first compact, modern-style thermometer with graduated

markings, and it was similar to today's devices. The Fahrenheit temperature scale (see 1740–1742) was named after him.

In Lisbon, meanwhile, Brazilian-born priest and naturalist **Bartholome de Gusmão** (1685–1724) sent a ball to the roof using **hot air** and designed a **hot-air airship**. Although the first recorded manned flight in a hot-air balloon would not happen for another 74 years, de Gusmão's experiment anticipated future developments in aviation.



Coral reefs may look like plants but really they are colonies of animals.

IN 1710, GERMAN PAINTER JACOB CHRISTOPH LE BLON

(1667–1741) found he could print pictures in a range of colors with just three different-colored inks. Paint of almost any color could be created by mixing three primary colors, but Le Blon realized that the colors did not have to be mixed. Instead, they could be **printed one on top of the other** in three layers. He started off in 1710 with three colors: red, blue, and yellow. Later, he discovered that even better results could be achieved with four colors: **black (K)** and the three primary colors used in printing—now known as **cyan (C), magenta (M), and yellow (Y)**, today called the CMYK system.

Also in this year, French entomologist **René de Réaumur** (1683–1757) set out to investigate whether spiders can make silk like silkworms. He showed that

17 CUBIC INCHES THE TOTAL CAPACITY OF THE HUMAN HEART

1707 John Floyer creates a watch for counting the pulse

1707 Herman Boerhaave introduces a systematic method for diagnosing illness

1708 Ehrenfried Walther von Tschirnhaus discovers the formula for making porcelain

1709 Gabriel Fahrenheit constructs an alcohol thermometer

January 10, 1709 Abraham Darby produces cast iron using coke

1709 Francis Hauksbee experiments with static electricity

August 8, 1709 Bartholome de Gusmão devises a hot-air airship

1710 Jacob Le Blon invents the three-color printing method

1710 René de Réaumur shows that spiders make silk